

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		award (2) for correct equation $^{124}_{53}\text{I} \rightarrow ^{124}_{52}\text{Te} + \beta^+$ if not correct award (1) for correct symbol for tellurium	2			2		
		(ii)		award (1) for points plotted correctly 50 units at 4 days, 25 units at 8 days, 12.5 units at 12 days and 6.25 units at 16 days award (1) for smooth curve through points	2			2	1	
	(b)			bromine has two isotopes ^{79}Br and ^{81}Br (1) m/z 158 due to molecules containing two of smaller isotope / ^{79}Br only, m/z 162 due to molecules containing two of larger isotope / ^{81}Br only and m/z 160 due to molecules containing one of each isotope / one ^{79}Br and one ^{81}Br (1) 50% of each isotope / ratio of isotopes is 1:1 therefore height of $^{158}\text{Br}_2^+$ and $^{162}\text{Br}_2^+$ same and $^{160}\text{Br}_2^+$ twice as big (1) accept answer in terms of probability	1		2	3		

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	(c)	(i)		$[H^+] = 1.20 \times 10^{-2}$		1		1	1	
		(ii)		moles of $Ba(OH)_2 = 1.18 \times 10^{-4}$ (1) moles of $H^+ = 2.36 \times 10^{-4}$ (1) $[H^+] = 9.46 \times 10^{-3}$ (1) ecf possible throughout		3		3	2	3
		(iii)		agree – since $[H^+]$ is different, the pH must be different			1	1		
				Question 10 total	5	4	3	12	4	3